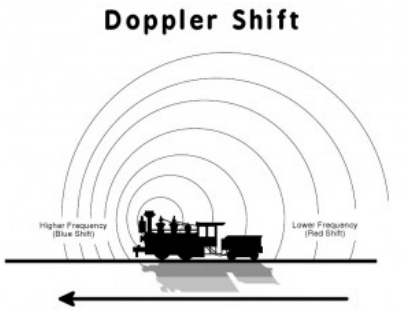
Redshift



https://www.youtube.com/watch?v=FhfnqboacV0

Total: /24

Watch the YouTube clip at the above address and answer the following questions.

I suggest you watch the entire clip at least once before you attempt to answer the questions!

1. What do astronomers use the term redshift to describe?

_____

_____ *how far a way a distant object is*

(1 mark)

2. What should you think about to understand what a 'cosmic redshift' is?

_____

_____ *sound*

(1 mark)

3. How does sound travel? _____ *waves*

(1 mark)

4. Why do you hear the engine of a plane?

_____

_____ *the waves travel through the air to your ears*

(1 mark)

5. If a plane is moving what happens to the sound waves in front of it?

_____

_____ *compressed*

(1 mark)

6. What happens to the sound? _____ *higher frequency*

(1 mark)

7. What happens to the sound waves behind the plane?

_____

_____ *stretched*

(1 mark)

8. What happens to the sound? _____ *lower frequency*

(1 mark)

9. What happens as the plane passes you?

_____

_____ *you hear the Doppler shift/effect*

(1 mark)

10. What happens to light waves if they compress or stretch?

_____

_____ *change frequency or colour*

(1 mark)

11. Describe the difference in wavelength between red and blue

_____

_____ *blue is shorter than red*

_____
_____

_____
_____

(1 mark)

12. Draw what happens to light waves as they travel towards you in space (label and colour)

(3 marks)

13. What does “blueshifted” mean?

_____

_____ *compressed, higher frequency, shorter wavelength. Need 2 for full marks*

(2 marks)

14. Draw what happens to light waves as they travel away from you in space (label and colour)

(3 marks)

15. What does ‘redshifted’ mean?

_____

_____ *stretched, lower frequency, longer wavelength. Need 2 for full marks*

(2 marks)

16. Why is the light from most galaxies that we see from Earth redshifted?

_____

_____ *Universe is expanding*

(1 mark)

17. The __ *farther*_____ away a galaxy is the _____ *faster*_____ its moving and the

__ *faster*_____ a galaxy moves the ___ *more*_____ its light is ___*redshifted*_____

So measuring the amount of ___ *redshift* _____ is a good measure of the ___ *distance*__

of the galaxy.

(2 marks)